

RFC-ALP-001: Agent Lexicon Protocol

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RFC-ALP-001: Agent Lexicon Protocol (ALP)

Status: Draft **Version:** 1.1-draft **License:** MIT

Changes from 1.0: Symbol identity is now derived from a composed primitive tree rather than from an opaque definition record (§3, §4). A closed primitive inventory (§4) and a written script (§6) are normative. Acquisition, competence and cold-start behavior change substantially as a consequence (§9, §11.1). A new failure mode, synonymy forking, is introduced and unsolved (§12.6).

Abstract

ALP is a protocol for machine agents to construct, exchange, and durably store a shared symbolic vocabulary that is denser than natural language, portable across tokenizers and model families, and interpretable independently of the models that produced it.

Meaning is **composed** from a closed inventory of roughly eighty semantic primitives. A symbol is a tree of primitives; its identifier is the hash of that tree. An agent holding the primitives can therefore decode a symbol it has never seen, by reading its composition. The lexicon is append-only and grows without coordination. Communication is a causally-ordered stream of immutable events, so receivers hold no session state and may join, leave, or be replaced mid-conversation.

A written script is defined so that composed symbols render as single visual blocks for human audit, in the manner of Hangul syllabic assembly.

1. Introduction

1.1 Problem Statement

Agents that converse in natural language pay for expressiveness they do not need. Agents that converse in a schema-bound binary format pay in brittleness: the schema becomes a coordination dependency, stored messages become unreadable without the registry live when they were written, and substituting one model for another invalidates the arrangement.

Four properties are required, and no existing serialization format supplies them jointly:

1. **Density.** Recurring concepts SHOULD cost a reference, not a restatement.
2. **Dynamism.** The vocabulary MUST grow mid-conversation, from either side, without a coordinator and without renumbering.
3. **Durability.** A stored conversation MUST remain interpretable after the models, the vocabulary, and the participants have all changed.
4. **Derivability.** A participant SHOULD be able to interpret a symbol it has never encountered, without a round trip.

Properties 1–3 follow from one decision: **a symbol's identity is the hash of its definition.** Property 4 follows from a second: **a definition is a composition of primitives, not opaque text.** Everything below is downstream of those two.

1.2 Non-Goals

ALP does not attempt to:

- Reduce the token cost of a model's own inference. At the boundary where a model ingests a conversation, ALP content is projected into a surface form that model reads well, and that projection costs approximately what natural language costs. ALP's savings are in transport, storage at rest, and selective hydration (§8.3) — not in the forward pass. **The script of §6 does not change this and MUST NOT be represented as compression** (§6.1).
- Guarantee semantic agreement. Composition narrows divergence sharply but does not eliminate it (§10).
- Replace a general-purpose serialization format.

1.3 Requirements Language

The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY, and OPTIONAL are to be interpreted as described in BCP 14 (RFC 2119, RFC 8174).

1.4 Terminology

Term	Definition
Agent	A participant in a stream. Backed by one or more models over its lifetime.
Primitive	An irreducible unit of meaning from the closed inventory (§4).
Symbol	A composed tree of primitives. Identified by SID.
SID	Symbol Identifier. Hash of the symbol's canonical composition tree.
Composition	The tree of primitives constituting a symbol's meaning.
Gloss	OPTIONAL natural-language text accompanying a composition. Advisory.
Lexicon	The set of symbols known to a stream. Append-only.
Event	An immutable frame in a stream. Identified by EID.
EID	Event Identifier. Hash of the canonical binary frame body.
Stream	A causally-ordered DAG of events constituting one conversation.
Competence	What an agent can correctly encode and decode. Primitive-level and symbol-level (§9).
Block	The rendered visual form of one symbol under the script (§6).
Hydration	Projecting ALP content into a surface form a model can reason over.

1.5 Prior Art

Design debts, stated so implementers can read the originals:

Source	Borrowed
Blissymbolics	Semantic composition from a bounded symbol set
Natural Semantic Metalanguage (Wierzbicka)	The claim that a small prime inventory suffices
Hangul	Assembly of a bounded inventory into visual blocks
ASN.1	Abstract syntax separated from transfer syntax
Git / Merkle DAG	Content-addressed identity, causal history
Avro	Writer-schema resolution, schema shipped with data
CRDT literature	Grow-only sets, strong eventual consistency
FIPA-ACL, KQML	Performatives, conversation identity

2. Architecture

2.1 Layer Separation

```
+-----+
| L4 Script      glyph blocks, font, typography |
+-----+
| L3 Competence  who holds which primitives    |
+-----+
| L2 Transfer    ALP/B binary | ALP/T canonical |
+-----+
| L1 Semantics   primitives, composition trees  |
+-----+
```

- **L1 (Semantics)** defines meaning: a closed primitive inventory and the composition rules over it.
- **L2 (Transfer)** defines encoding. ALP/B for transport, ALP/T for storage, audit and external translation. Both MUST be lossless with respect to L1 and MUST round-trip to each other.
- **L3 (Competence)** defines capability.
- **L4 (Script)** defines rendering. It is **non-normative for meaning**. An implementation MUST NOT allow L4 to influence an L1 identity, and a stream MUST be fully functional with L4 absent.

Hashes are computed over canonical ALP/B of L1 structures only. ALP/T and the script are projections and are never authoritative.

2.2 Event Stream Model

A conversation is not a session. It is an append-only log of immutable events forming a causal DAG. Consequences:

- Receivers are stateless. All state is a deterministic fold over the event set (§7.4).
- An agent MAY join mid-stream, leave, and rejoin. There is no resume handshake.
- An agent MAY replace its backing model between any two events (§9.4).
- Two agents MAY emit concurrently without a coordinator. Concurrency is represented, not prevented (§7.2).
- Replay is idempotent. Events are content-addressed, so applying one twice is a no-op.

3. Symbol Identity

3.1 Composition Record

A symbol is defined by a tree. The tree is the definition; there is nothing else authoritative about the symbol.

Field	Key	Type	Req	Description
Head	<code>h</code>	PID	MUST	The governing primitive (§4.2)
Roles	<code>r</code>	MAP<u8, Node>	MAY	Ordered semantic roles, keyed by role code (§5.3)
Modifiers	<code>m</code>	SET<Node>	MAY	Unordered qualifiers. Canonically sorted (§3.3).
Supersedes	<code>s</code>	SID	MAY	Prior symbol this refines (§3.5)
Gloss	<code>g</code>	text	MAY	Advisory only. Excluded from the hash (§3.4).
Residue	<code>x</code>	text	MAY	Meaning not expressible in primitives (§5.5). Included in the hash.

A `Node` is recursively either a PID, a nested Composition Record, or a SID reference to an existing symbol.

Gloss is excluded from the hash and Residue is included. This is deliberate: gloss is a convenience for humans and MUST NOT be able to fork two otherwise-identical symbols, whereas residue is load-bearing meaning that composition failed to capture, and two symbols differing in residue are different symbols.

3.2 Label Removal

Version 1.0 carried a non-identifying `Label` field. It is removed. A symbol's human-facing name is now derivable from its composition, and any additional naming belongs in tooling, not in the record.

3.3 Canonical Composition

Hashing requires exactly one tree serialization per meaning.

- `Roles` is a map keyed by role code and MUST be serialized in ascending role-code order. Role position is semantic; it MUST NOT be sorted by content.
- `Modifiers` is a set. Its members MUST be serialized in ascending order of their canonical encoded bytes. Duplicate members MUST be removed.
- Nested Composition Records are canonicalized by the same rules, recursively.
- Absent OPTIONAL fields MUST be omitted, never encoded as null.
- `Gloss` MUST be stripped before hashing and MAY be retained in transport.

Rule 1 versus rule 2 is the crux: ordered roles preserve argument structure (`A prevents B` differs from `B prevents A`), while unordered modifiers converge (`HIGH then FUTURE` is the same symbol as `FUTURE then HIGH`). Getting this wrong produces silent lexicon forks.

3.4 SID Computation

```
SID = H( canonical_ALP/B( CompositionRecord \ gloss ) )
```

`H` is SHA-256 by default. Implementations MUST support SHA-256, MAY negotiate others, and MUST NOT vary the algorithm within a stream.

Because a SID is derived from composition and nothing else:

- No authority assigns identifiers. There is no registry to be unavailable, partitioned, or shut down.
- Two agents composing the same meaning converge on the same SID automatically, without ever having communicated.
- Concurrent amendment is not a conflict. Leader election, merge resolution and renumbering are structurally impossible to need.
- A symbol is self-authenticating: its meaning is inside the hash preimage, so a supplied composition either matches its SID or is rejected (§11.3).
- A symbol is derivable.** An agent holding the primitive inventory can read a composition it has never seen. This is what §1.1 property 4 buys, and it is new in 1.1.

3.5 Truncation Profiles

Profile	Bytes	Collision risk (birthday)	Use
SID-256	32	negligible	Archival, adversarial, cross-org
SID-128	16	$\sim 2^{64}$ symbols	RECOMMENDED default
SID-96	12	$\sim 2^{48}$ symbols	High-volume trusted transport
SID-64	8	$\sim 2^{32}$ symbols	NOT RECOMMENDED

Archived streams **MUST** store SIDs at `SID-256` regardless of transport profile. Truncation is a transport optimization and **MUST NOT** propagate into storage.

3.6 Supersession, Not Redefinition

A symbol’s meaning is immutable. Changing a meaning means composing a new symbol whose `supersedes` points at the old SID.

This is what makes epoch markers unnecessary. Under a mutable vocabulary a message’s meaning depends on when it is read, so every stored message needs a point-in-time registry lookup and every in-flight message is ambiguous during renegotiation. Under ALP a message’s meaning is fixed at write time by construction. An event from turn one reads identically after ten thousand amendments.

Supersession is advisory. A receiver **MAY** continue using a superseded symbol and **MUST** still decode it.

4. Primitive Inventory

4.1 Closure

The inventory is **closed**. It **MUST NOT** be extended within a stream, and extension between versions is a protocol revision requiring a new version byte.

Closure is what makes derivability work: an agent that has acquired the inventory once can read anything, forever, from anyone. An open inventory collapses this back to the 1.0 behavior of learn-on-demand.

The cost is a hard expressiveness ceiling — Blissymbolics hit exactly this wall. §5.5 defines the escape hatch, and §12.2 records the tradeoff.

4.2 Classes and Codes

Primitive Identifiers (PIDs) are two-byte codes. The high byte is the class, the low byte the member. Codes are also the Unicode PUA offsets used by the script (§6.3).

Class 0x00 — Ontological. What kind of thing this is. Every composition **MUST** have an ontological head.

PID	Name	Sense
0x0000	ENTITY	A thing that persists
0x0001	PROCESS	Something that unfolds
0x0002	PROPERTY	An attribute borne by something
0x0003	RELATION	A tie between two or more things
0x0004	QUANTITY	A measurable magnitude
0x0005	AGENT	An entity capable of action
0x0006	STATE	A condition holding at a time
0x0007	PLACE	A location or region
0x0008	MOMENT	A time or interval as an object
0x0009	SIGN	Information, a message, a representation
0x000A	EVENT	A bounded occurrence
0x000B	GROUP	A set or collection

Class 0x01 — Modal.

PID	Name	Sense
0x0100	AFFIRM	Holds
0x0101	NEGATE	Does not hold
0x0102	POSSIBLE	May hold
0x0103	NECESSARY	Must hold
0x0104	DESIRED	Wanted to hold
0x0105	HYPOTHETICAL	Entertained, not asserted
0x0106	PERMITTED	Allowed to hold
0x0107	FORBIDDEN	Not allowed to hold

Class 0x02 — Scalar.

PID	Name	Sense
0x0200	NONE	Zero
0x0201	SOME	Nonzero, partial
0x0202	ALL	Total
0x0203	LOW	Low on the relevant scale
0x0204	MID	Middling
0x0205	HIGH	High
0x0206	EXTREME	At or beyond the limit
0x0207	BOUNDED	Has limits
0x0208	UNBOUNDED	Has none
0x0209	INCREASE	Rising
0x020A	DECREASE	Falling

Class 0x03 — Temporal.

PID	Name	Sense
0x0300	PAST	Before now
0x0301	NOW	At present
0x0302	FUTURE	After now
0x0303	DURATIVE	Extended in time
0x0304	PUNCTUAL	Instantaneous
0x0305	BEFORE	Prior to a reference
0x0306	DURING	Concurrent with a reference
0x0307	AFTER	Subsequent to a reference
0x0308	REPEAT	Recurring
0x0309	BEGIN	Onset
0x030A	END	Cessation

Class 0x04 — Causal.

PID	Name	Sense
0x0400	CAUSE	Brings about
0x0401	ENABLE	Makes possible
0x0402	PREVENT	Makes impossible
0x0403	CORRELATE	Co-varies, causation unclaimed
0x0404	DEPEND	Requires as precondition
0x0405	TRIGGER	Initiates on occurrence

Class 0x05 — Epistemic.

PID	Name	Sense
0x0500	KNOWN	Established
0x0501	BELIEVED	Held without proof
0x0502	INFERRED	Derived by reasoning
0x0503	UNKNOWN	Not established
0x0504	CONTESTED	Disputed between parties
0x0505	OBSERVED	Directly witnessed
0x0506	PREDICTED	Projected forward

Class 0x06 — Illocutionary. Present in a composition only when the symbol names a speech act; the event type (§7.1)

governs the frame.

PID	Name	Sense
0x0600	ASSERT	Puts forward as true
0x0601	REQUEST	Asks for action
0x0602	COMMIT	Binds the speaker
0x0603	QUERY	Asks for information
0x0604	WARN	Flags hazard
0x0605	REFUSE	Declines
0x0606	PROPOSE	Offers for consideration
0x0607	ACKNOWLEDGE	Registers receipt

Class 0x07 — Valence and Deontic.

PID	Name	Sense
0x0700	GOOD	Positively valued
0x0701	BAD	Negatively valued
0x0702	REQUIRED	Obligatory
0x0703	OPTIONAL	Discretionary
0x0704	SAFE	Without hazard
0x0705	HARM	Damage or injury
0x0706	COST	Expenditure
0x0707	BENEFIT	Gain

Class 0x08 — Structural. Grammar, not semantics. These MUST NOT appear as an ontological head.

PID	Name	Sense
0x0800	REF	Points at an existing SID
0x0801	SCOPE_OPEN	Begins a nested composition
0x0802	SCOPE_CLOSE	Ends one
0x0803	SUPERSEDE	Marks the supersession link
0x0804	RESIDUE	Marks untranslated meaning (§5.5)

Total: 76 primitives across 9 classes, 52 codes reserved per class for revision.

5. Grammar

5.1 Well-Formedness

A Composition Record is well-formed iff:

1. `Head` is a class 0x00 primitive.
2. No class 0x08 primitive appears as a head or modifier.
3. Nesting depth does not exceed 8. Implementations MUST reject deeper trees with `E_DEPTH`.
4. Modifier count per node does not exceed 12.
5. It contains no cycle. Since SIDs are content-derived, a symbol cannot reference itself or any ancestor; implementations MUST reject with `E_CYCLE`.

Constraints 3 and 4 exist to bound the script's block complexity (§6.4) and to make denial of service by pathological composition impractical.

5.2 Modifier Semantics

Modifiers attach to their node and qualify it. Modification is commutative and idempotent — `HIGH` applied twice is `HIGH` once, which is why modifiers are a set. Modifiers from the same class MAY co-occur but SHOULD NOT be contradictory; `AFFIRM` with `NEGATE` on the same node is well-formed and meaningless, and implementations SHOULD warn.

5.3 Roles

Code	Role	Sense
0x01	ARG0	Agent, cause, or first argument
0x02	ARG1	Patient, effect, or second argument
0x03	ARG2	Instrument, recipient, third argument
0x04	SCOPE	The domain over which the head applies
0x05	MEASURE	Unit or scale reference
0x06	CONDITION	Precondition on the head

Roles are ordered by code and are semantically positional. `RELATION` with head `PREVENT`, `ARG0=deploy`, `ARG1=outage` is a different symbol from the same with the arguments swapped, and both hash differently. This is the intended behavior.

5.4 Value Binding

A symbol names a concept; it does not carry data. Data is bound at assertion time in an `ASSERT` payload as `(SID, Value)` pairs (§7.1). A composition MAY constrain admissible values via a `MEASURE` role, but constraint enforcement is an implementation concern and is not specified here.

5.5 Residue

Meaning that does not decompose is carried in the `Residue` field as natural language, marked by `0x0804`. Residue is included in the hash and is therefore load-bearing.

Residue is the pressure valve for §4.1 closure. It MUST be usable, and it SHOULD be rare: a lexicon in which most symbols carry residue has degraded to version 1.0 behavior and loses derivability. Implementations SHOULD report residue rate as an operational metric.

6. The Script

6.1 Purpose and Explicit Non-Purpose

The script renders a composition as a single visual block, so a human auditing a stored stream sees structured glyphs rather than hashes.

It does not reduce token cost. Private Use Area codepoints are outside model tokenizer vocabularies and fall through to byte-level fallback, typically three to four tokens per glyph — worse than ASCII, and models reason poorly over them. The script buys human audit, visual scanning, diff legibility and an identity for the language. It buys nothing at the model boundary, and implementations MUST NOT present it as compression.

Agents MAY emit script for human-facing output. Agents SHOULD NOT use it as an inter-agent transfer syntax; ALP/B exists for that.

6.2 Assembly Model

Assembly follows Hangul: a bounded inventory of parts composes into a block whose internal arrangement is meaningful. The head occupies the block center at full weight. Modifiers attach as marks in class-assigned positions:

```

      [ modal | epistemic ]
    [ causal ] --- HEAD --- [ scalar ]
      [ temporal | valence ]

roles descend beneath the block as an ordered stack
```

Position encodes class, so a reader can identify what kind of qualification is present before decoding which one. A block is legible at a glance for structure, and on inspection for content.

6.3 Codepoints

Primitives occupy the Basic Multilingual Plane PUA at `U+E000 + PID`, so `ENTITY` is `U+E000`, `NEGATE` is `U+E101`, `REF` is `U+E800`. Class `0x08` structural primitives are format controls and SHOULD render as zero-width or as bracket forms.

Supplementary Plane 15 (`U+F0000–U+FFFFD`) is reserved for a future precomposed block range if profiling shows shaping cost matters.

6.4 Font Requirements

A conforming font MUST:

1. Provide a base glyph for every PID in §4.2.
2. Implement `ccmp` and `GSUB` such that a canonical primitive sequence forms one block.
3. Implement `GPOS` mark attachment for the positions of §6.2.
4. Degrade to a legible linear sequence of base glyphs when shaping is unavailable.
5. Supply `.notdef` handling that renders unknown PUA codepoints visibly rather than silently.

Requirement 4 matters more than it looks: a great deal of tooling will not shape, and a stream that becomes unreadable in `grep` output has lost the audit property the script exists to provide.

Reference toolchain on Fedora: `fontforge`, `python-fonttools`, `harfbuzz`.

6.5 Fallback Transliteration

Every composition MUST have an ASCII transliteration, since the script is OPTIONAL and ALP/T MUST remain greppable. The form is the ALP/T composition syntax of §7.6, which is normative; the script is a rendering of it.

7. Wire and Text Formats

7.1 Event Types

Type	Code	Purpose
JOIN	0x01	Announce participation and competence
LEAVE	0x02	Announce departure. Advisory.
ASSERT	0x03	State content using lexicon symbols
AMEND	0x04	Introduce Composition Records into the lexicon
EXPAND	0x05	“I do not hold SID X; supply its composition”
GROUND	0x06	Supply Composition Records for requested SIDs
REGROUND	0x07	“I believe we have diverged on SID X” (§10)
ACQUIRE	0x08	Offer inventory or vocabulary to a joining agent
ATTEST	0x09	Claim or demonstrate competence
CHECKPOINT	0x0A	Materialized state snapshot for fast join
ERROR	0x0B	Signal a protocol fault (§11.5)

EXPAND and GROUND remain, but their expected frequency drops sharply under composition: they are needed only when a receiver lacks the primitive inventory or encounters residue it cannot interpret.

There is no ACK. Acknowledgment is implicit in causal reference.

7.2 Causality

`parents` holds the EIDs of every event the author had observed and not yet referenced, at emission. This yields a Merkle DAG with Git’s properties: partial order without a clock, tamper evidence, lossless concurrency.

- Events with no causal path between them are concurrent. This is legal and is NOT an error.
- Where a total order is required, implementations MUST use the causal partial order, breaking ties by bitwise-ascending EID. Deterministic and identical across agents.
- `timestamp` is advisory and MUST NOT be used for ordering.

7.3 CHECKPOINT

Field	Type	Description
<code>covers</code>	LIST<EID>	DAG frontier summarized
<code>hash_alg</code>	u8	Stream hash algorithm
<code>sid_profile</code>	u8	Truncation profile (§3.5)
<code>inventory_ver</code>	u8	Primitive inventory version (§4.1)
<code>lexicon</code>	LIST<SID>	Live symbols
<code>comps</code>	LIST<CompRec>	OPTIONAL inline compositions for the hot subset
<code>state</code>	MAP	Materialized assertions
<code>digest</code>	BYTES	Hash of canonical <code>state</code>

Any agent MAY emit a CHECKPOINT. A receiver MUST verify `digest` independently before trusting one from an

untrusted peer. Two correct agents checkpointing the same frontier MUST produce identical digests; mismatch MUST raise `E_DIVERGENCE`.

`inventory_ver` is new in 1.1 and MUST match across participants. A mismatch is fatal to derivability and MUST raise `E_INVENTORY`.

7.4 Stateless Fold

```
state(E) = fold(apply, S0, toposort(E))

S = (L, C, A)
L : SID -> CompositionRecord    ; lexicon,    grow-only
C : AGT -> Set<PID | SID>       ; competence, grow-only per agent
A : SID -> Value                ; assertions, last-writer-wins
```

`L` and `C` are grow-only sets. Union is commutative, associative and idempotent, so they form a join-semilattice and converge under strong eventual consistency with no coordination. **This is why append-only is required rather than merely convenient** — it is the property that lets agents amend the vocabulary concurrently and still agree.

`A` resolves conflicting concurrent assertions by the total order of §7.2. Implementations requiring different merge semantics MUST declare them at CHECKPOINT.

`apply` MUST be idempotent; re-applying a seen EID is a no-op.

7.5 ALP/B

ALP/B uses a tag-length-value encoding intentionally shaped like CBOR (RFC 8949) so existing decoders can be adapted. It is not CBOR-compatible; major type assignments differ.

A tag byte is `MMMMAAAA`: 3-bit major type, 5-bit argument. Arguments 0-23 are immediate; 24-27 indicate a following 1, 2, 4 or 8-byte big-endian argument; 31 indicates indefinite length terminated by `0xFF`.

Major	Type	Argument
0	UINT	the value
1	NINT	-(value + 1)
2	BYTES	byte count
3	TEXT	byte count
4	LIST	element count
5	MAP	pair count
6	REF	reference kind
7	SIMPLE	false=20, true=21, null=22, f64=27

REF arguments: 0 = SID at stream profile, 1 = EID at stream profile, 2 = SID full 32 bytes, 3 = EID full 32 bytes, **4 = PID (two bytes, new in 1.1)**.

Canonical form requires: shortest integer encoding; no indefinite length; map keys sorted bitwise ascending; no duplicate keys; absent optionals omitted; f64 floats with canonical quiet NaN; NFC-normalized UTF-8. Non-canonical bytes where canonical form is required MUST be rejected with `E_NONCANONICAL`.

Frame:

```
stream := frame*
frame  := length:uvarint body:bytes[length]

body   := version:u8 type:u8 flags:u8
         stream_id:REF(EID)
         parents:LIST<REF(EID)>
         timestamp:UINT
         author:REF(SID)
         payload:<type-specific>
         [signature:BYTES] ; iff flags & 0x01

EID    := H( canonical_ALP/B( body ) ) ; signature excluded
```

Flag	Bit	Meaning
SIGNED	0x01	Trailing detached signature over body sans signature
CHECKPOINT_REF	0x02	Payload carries a checkpoint EID for fast join
EPHEMERAL	0x04	Sender requests exclusion from archive
HYDRATE_HINT	0x08	Payload carries a projection hint (§8.3)

Excluding the signature from EID means signed and unsigned copies share identity and do not fork the DAG.

7.6 ALP/T

ALP/T is REQUIRED for archival, OPTIONAL for transport. It MUST round-trip to byte-identical canonical ALP/B. Hashes are never computed over it.

```
document := "%alp/t" SP version SP "inv" SP inv_ver NL block+
block    := event | symbol
symbol   := "!" sid SP comp NL ("=" SP quoted NL)?
comp     := head modifier* role* residue?
head     := prim
modifier := "." prim
role     := ":" rolename SP (prim | "(" comp ")" | "#" sid)
residue  := "~" quoted
prim     := "$" primname
event    := "@" eid SP type NL
          ("<" SP eid (SP eid)* NL)?
          ("by" SP "#" sid NL)
          ("at" SP iso8601 NL)
          (INDENT term NL)+
```

The = "text" line is the gloss. It is excluded from the hash (§3.1) and is present purely so an archive reads well.

8. Density and Hydration

8.1 Where the Savings Are

Path	ALP effect
Agent-to-agent transport	Large reduction. Symbols are references.
Storage at rest	Large reduction, plus interpretability guarantee.
Selective hydration	Large reduction. Dominant win for long-running agents.
Cold start	Improved over 1.0. Composition is derivable (§11.1).
Model ingestion of hydrated content	Approximately neutral.
Script rendering	Negative at the model boundary (§6.1).

8.2 First-Use Cost

Under 1.0, first use of a symbol cost an EXPAND/GROUND round trip. Under composition it usually costs nothing: a receiver holding the inventory decodes the composition inline. The remaining round-trip cases are missing inventory (once per agent lifetime, not per symbol) and uninterpretable residue.

8.3 Selective Hydration

A model does not need the whole transcript, only the symbols relevant to its task. Because the fold is pure and assertions are keyed by SID, an agent MAY hydrate an arbitrary subset:

```
hydrate(state, sids, competence) -> surface_form
```

For a long stream this is the difference between replaying ten thousand events and materializing forty assertions. Implementations SHOULD expose hydration scope explicitly rather than defaulting to full replay.

Hydrated surface form is generated output. It MUST NOT re-enter the stream as authoritative and MUST NOT participate in any hash. An agent that has reasoned over hydrated content and wishes to record a conclusion MUST emit a new ASSERT.

9. Competence

9.1 Two Levels

Competence is now stratified, which is the largest practical change from 1.0.

- **Primitive competence** is the inventory of \$4. It is finite, closed and acquired once. An agent either holds it or does not.
- **Symbol competence** is derived. An agent with primitive competence has presumptive competence over every symbol composed solely from primitives, without transmission.

The consequence: the $O(\text{agents} \times \text{symbols})$ competence tracking of version 1.0 is mostly gone. A sender needs to track

only whether a receiver holds the inventory, plus the residue-bearing exceptions.

9.2 ATTEST

Level	Code	Meaning
HELD	0x01	Stored. Decode claimed, not demonstrated.
DEMONSTRATED	0x02	Passed a round-trip challenge.
DECLINED	0x03	Cannot or will not work this. Do not send unexpanded.

ATTEST now carries PIDs as well as SIDs. An agent SHOULD attest the inventory at JOIN and thereafter attest only exceptions.

DEMONSTRATED proves correct transformation. **It does not prove semantic agreement** (§10).

9.3 Unknown Symbols Never Fail a Message

On encountering a SID it cannot resolve, a receiver MUST NOT discard the event. It MUST emit EXPAND naming the unresolved SIDs and SHOULD buffer the event pending GROUND.

Failing loses the message permanently and demands a perfect competence model. Expansion costs one transmission, amortized across every later use. A wrong competence estimate is therefore a performance problem, never a correctness one.

9.4 Model Substitution

An agent MAY change its backing model between any two events. No session state exists to be lost.

The incoming model must nonetheless acquire the language. Under 1.1 this is bounded work: acquire the closed inventory of \$4 once, and every symbol in the stream — past, present and future — becomes readable. The substituting agent MUST:

1. Acquire the primitive inventory and verify `inventory_ver` against the current CHECKPOINT.
2. OPTIONALLY demonstrate competence by round-trip challenge over a sample.
3. Emit ATTEST reflecting actual competence, including DECLINED for anything it cannot work.

Partial competence remains representable and expected. Implementations MUST NOT treat acquisition as pass/fail. Competence is grow-only within the fold, so DECLINED is recorded as a distinct assertion rather than a retraction.

10. Semantic Drift

Composition narrows divergence but does not close it, and implementers should not be misled about the residual.

Passing a round-trip challenge proves an agent decodes a symbol. It does not prove the agent attaches the same meaning. Composition helps materially: `PROPERTY.HIGH.PUNCTUAL:MEASURE(...)` constrains interpretation far more than a paragraph of grounding text, because the constraint is structural rather than linguistic. But HIGH on a latency scale is still a judgment, and two agents can hold different thresholds while composing identically.

Residue (§5.5) reintroduces the full 1.0 problem wherever it appears, since residue is natural language by definition.

Human language has this failure and survives it through **repair**. ALP makes repair first-class:

Field	Type	Description
subject	SID	Symbol suspected of divergence
evidence	LIST<EID>	Events exhibiting the mismatch
reading	text	The sender's operative interpretation
proposal	SID	OPTIONAL newly composed superseding symbol

REGROUND is not an error. It is expected in a long conversation and implementations SHOULD surface rather than suppress it. A stream with no REGROUND over thousands of events is more likely drift-accumulating than drift-free.

Detection remains unspecified.

11. Security Considerations

11.1 Truncation Collisions

SID-64 and SID-96 are attackable by an adversary who can compose symbols: a collision lets a malicious composition impersonate a trusted one. Streams with untrusted participants MUST use SID-128 or wider and SHOULD use SID-256.

11.2 Lexicon Growth as Denial of Service

The lexicon is grow-only and composition is cheap. Implementations MUST rate-limit AMEND per author and SHOULD cap lexicon size, emitting E_LEXICON_FULL rather than degrading.

Composition adds a second vector: deeply nested or highly modified trees are expensive to canonicalize and render. §5.1 bounds this at depth 8 and 12 modifiers per node; implementations MUST enforce both.

11.3 Composition Poisoning

Compositions are self-authenticating: SID is the hash of the composition, so a receiver MUST verify supplied compositions hash to the requested SID and MUST reject mismatches with E_SID_MISMATCH.

An attacker cannot alter an existing symbol's meaning. It can compose a different symbol with misleading structure and induce its use, which is detectable at the SID level.

New in 1.1: because gloss is excluded from the hash, an attacker MAY attach a misleading gloss to a legitimate SID without breaking verification. Receivers MUST treat gloss as advisory and MUST resolve meaning from composition. Implementations SHOULD warn where a received gloss differs from a locally stored one for the same SID.

11.4 EXPAND Amplification

Implementations SHOULD rate-limit GROUND per requester and MAY require ATTEST before honoring bulk EXPAND.

11.5 Error Codes

Code	Name	Meaning
0x01	E_VERSION	Unsupported protocol version
0x02	E_NONCANONICAL	Canonical encoding violated
0x03	E_SID_MISMATCH	Composition does not hash to claimed SID
0x04	E_UNKNOWN_PARENT	Referenced parent EID unresolvable
0x05	E_LEXICON_FULL	Lexicon cap reached
0x06	E_RATE	Rate limit exceeded
0x07	E_DIVERGENCE	Checkpoint digest mismatch
0x08	E_PROFILE	SID profile mismatch within stream
0x09	E_INVENTORY	Primitive inventory version mismatch
0x0A	E_DEPTH	Composition nesting or modifier bound exceeded
0x0B	E_CYCLE	Cyclic composition
0x0C	E_MALFORMED	Well-formedness violated (§5.1)

11.6 Authenticity

Signing is OPTIONAL. Where participants are mutually untrusted, events SHOULD set SIGNED and carry a detached signature over the canonical body. Replay is not a threat: events are content-addressed and application is idempotent.

12. Tradeoffs

Presented for decision, not resolved here.

12.1 Inventory Size

A small inventory is fast to acquire, easy to render and easy to agree on, but forces heavy nesting and frequent residue. A large inventory expresses more directly but is slower to acquire, harder to render distinctly and more likely to contain near-synonyms that fork the lexicon (§12.6). The 76 of §4.2 is a starting point, not a result. **Decision:** inventory size and whether to profile-split it by domain.

12.2 Closure

Closure buys derivability (§4.1); it costs expressiveness. Residue mitigates but degrades toward 1.0 behavior in

proportion to its use. **Decision:** hard closure with residue, versioned inventory extension, or per-domain inventory profiles negotiated at CHECKPOINT.

12.3 Checkpoint Frequency

Frequent CHECKPOINT means fast joins and cheap model substitution at the cost of stream volume; checkpoints are the largest events in the protocol. **Decision:** fixed interval, size-triggered, or on-demand.

12.4 Script Investment

The script is a substantial build — glyph design for 76 primitives, GSUB/GPOS shaping, fallback behavior — and returns human auditability only. It costs tokens at the model boundary rather than saving them. **Decision:** whether the audit value justifies the build now or after the protocol is proven.

12.5 Compaction

The lexicon only grows. Compaction is the one operation that reintroduces the resolvability problem the design eliminates: dropping a symbol renders every historical event using it uninterpretable. Composition softens this — a dropped symbol is re-derivable from its composition if that composition is retained — but does not remove it. **Decision:** never compact; compact live state with a cold archive; or accept lossy history beyond a horizon.

12.6 Synonymy Forking — new in 1.1

Two agents can compose genuinely different primitive trees for the same concept. Both are well-formed, both hash differently, and the lexicon forks permanently with no automatic means of noticing. This is the cost of removing the coordinator, and it is more likely than the 1.0 hash-collision risk it replaces.

Mitigations, none complete: canonicalization rules that collapse provably-equivalent trees; a convention layer of blessed compositions for common concepts, which reintroduces coordination; or accepting the fork and relying on REGROUND to notice. **Decision:** required before implementation.

13. Open Issues

1. **Drift detection** (§10) has no specified mechanism.
2. **Synonymy forking** (§12.6) has no automatic detection.
3. **Compaction** (§12.5) has no safe formulation.
4. **Cross-stream reuse.** SIDs are globally unique by construction and compositions are self-describing, so symbols are trivially portable between streams. No transfer mechanism is specified.
5. **Competence transitivity.** If A trusts B’s ATTEST and B trusts C’s, A has no basis to trust C’s. Undefined.
6. **Inventory adequacy.** Whether 76 primitives cover the intended domains is an empirical question. Residue rate (§5.5) is the metric to instrument.

Appendix A: Worked Compositions

Transliteration per §7.6. \$ marks a primitive, . a modifier, : a role.

Concept	Composition	Reading
urgency	\$PROPERTY . HIGH . PUNCTUAL . REQUIRED	A property, high on its scale, instantaneous, obligatory
deadline	\$MOMENT . FUTURE . PUNCTUAL . BOUNDED	A time object, ahead, a point, limited
outage	\$STATE . NEGATE . NOW . BAD : SCOPE \$PROCESS	A present state of a process not holding, negatively valued
deprecated	\$PROPERTY . PAST . END : SCOPE (REF #a91c)	A property of having ended, scoped to a referenced symbol
escalate	\$PROCESS . INCREASE . FUTURE : ARG0 \$AGENT : ARG1 \$SIGN	An agent raising a signal, forward in time
suspected root cause	\$RELATION : ARG0 \$EVENT : ARG1 \$EVENT . CAUSE . INFERRED . CONTESTED	An inferred, disputed causal tie between two events
rollback permitted	\$PROCESS . PAST . PERMITTED : ARG1 \$STATE	Returning to a prior state is allowed
unknown blast radius	\$QUANTITY . UNBOUNDED . UNKNOWN : SCOPE \$GROUP : MEASURE \$ENTITY	An unbounded, unestablished count over a group of entities

Note that `urgency` and `deadline` share no primitives and are structurally distinguishable at a glance under the script. Note also that `escalate` and `suspected root cause` demonstrate ordered roles: swapping `ARG0` and `ARG1` yields a different SID, as §5.3 requires.

Appendix B: Minimal Exchange

Agent B has just replaced its model and holds the inventory but no vocabulary.

```
%alp/t 1 inv 1

@0001 JOIN          by #b000 at 2026-08-31T14:00:00Z
  (attest inventory 1 DEMONSTRATED)

@0002 ASSERT        by #a000 at 2026-08-31T14:00:02Z
<- 0001
  (#88b1 4)

@0003 AMEND          by #a000 at 2026-08-31T14:00:02Z
<- 0002
  !88b1c07d $PROPERTY.HIGH.PUNCTUAL.REQUIRED
  = "Operator priority 0-4. 4 pages on-call immediately."
```

B decodes `#88b1` from its composition on receipt of `0003` with no EXPAND round trip. Under version 1.0 this exchange required four events and a full grounding transmission; the AMEND here is 22 bytes of composition at `SID-128`. Had `0003` preceded `0002`, no additional event would have been needed at all.

14. Errata and revision plan

Recorded after external review of the 1.1 draft. These are accepted defects, not open questions. They are listed here rather than silently fixed because each implies a structural change that has not yet been approved.

14.1 ALP/B should be DAG-CBOR

Section 7.5 specifies a TLV encoding described as “intentionally shaped like CBOR” and then assigns incompatible major types. This is the worst available option: it forfeits the existing CBOR ecosystem while adding no capability. DAG-CBOR already provides strict canonical form and content-addressed links, which is precisely what sections 3.3, 3.4 and 7.5 exist to specify.

Revision: delete section 7.5 and normatively reference DAG-CBOR, adding one tagged type for PIDs. Section 3.5 truncation profiles should go with it, subsumed by CID multihash. The reference implementation in Appendix E hand-rolls the encoder only so that SID derivation is auditable without a dependency; it should not survive into an implementation.

14.2 The causal DAG and CRDT layer are reinvented

Sections 7.2 and 7.4 restate work already done correctly elsewhere. The sync protocol and DAG storage of a mature CRDT library should be adopted rather than specified here.

One caveat against wholesale adoption: mainstream CRDT document models offer mutable keys with per-key last-writer-wins, whereas the lexicon here is a grow-only set. A G-Set is strictly simpler and needs none of that machinery. The borrowing should be of transport and storage, not of document semantics.

Revision: retain the fold of section 7.4 as the specification of state, delegate its transport.

14.3 The novelty claim in section 1.1 is overstated

Section 1.1 asserts that no existing format satisfies the four properties jointly. Content-addressed IPLD plus a CRDT plus a sync protocol covers properties 1 through 3. The defensible claim is narrower: that these components have not been assembled around a closed composed semantic inventory, which is what supplies property 4.

Revision: restate section 1.1 as an assembly claim.

14.4 The authoritative meaning is still English

Every primitive in section 4.2 is defined by an English sense column, and residue under section 5.5 is English verbatim. Composition does not escape natural language; it relocates the dependency.

This cannot be fixed, and the specification should say so rather than imply otherwise. No formal system grounds its own primitives; NSM is explicit about exactly this. What composition does buy is a change of quantity that matters

operationally: English goes from unbounded, a fresh prose grounding per symbol forever, to 76 fixed definitions agreed once. A bounded, auditable, acquire-once dependency is a materially different thing from an open-ended one, but it is not independence.

A stronger form worth investigating: ground each primitive behaviourally, by a set of discrimination tests an agent must pass, rather than lexically by a gloss. This does not escape grounding either. It moves the dependency from agreement on an English sentence to agreement on a set of judgments, which is at least testable.

Revision: state the limit plainly in section 1.1, and add residue rate as a required operational metric.

14.5 What survives review

The primitive inventory and composition-as-identity (sections 3 to 5), competence stratification (section 9), and repair as a first-class message (section 10). Everything else is assembly over proven infrastructure, which is the correct outcome: a small delta over existing systems is a shorter path to a working prototype than a protocol specified from first principles.

Appendix D: A complete worked conversation

A two-agent incident exchange in ALP/T. Every event type is exercised, including a concurrent fork, a repair, and a mid-stream model substitution.

Every SID and EID below was computed by the reference implementation of Appendix E, not written by hand. Lines beginning with a semicolon are commentary and are not part of the format.

Five things to watch for while reading:

1. At event 3, `b000` receives two symbols it has never seen and needs no round trip. It reads them from their compositions. Under version 1.0 this required an EXPAND/GROUND exchange per symbol.
2. At events 8 and 9, two AMEND events name the same parent. Neither agent observed the other. Both are legal, both hash distinctly, and the fork is merged at event 10 by an event naming both as parents. No coordinator was involved at any point.
3. Those same two events differ only in the order of `ARG0` and `ARG1`, and produce different SIDs. Roles are positional; modifiers are not.
4. At event 12, both agents decode the urgency symbol correctly and still disagree about what it means. That is the residual problem of section 10. Event 13 supersedes rather than redefines, so event 11 remains unambiguous.
5. At event 16, an agent replaces its backing model. It acquires the closed inventory once and the entire stream becomes readable. The single EXPAND that follows at event 17 is for a symbol carrying residue, which is the one thing composition cannot derive.

```
%alp/t 1 inv 1

; a000 joins holding the closed inventory. Note it attests the inventory, not a vocabulary.
@9220b68c JOIN
by #a000
at 2026-08-31T14:20:00Z
  attest inventory 1 DEMONSTRATED

@c2affd08 JOIN
<- 9220b68c
by #b000
at 2026-08-31T14:20:02Z
  attest inventory 1 DEMONSTRATED

; Two symbols minted. b000 has never seen either and needs no round trip: it reads them from the composition.
@d77151b AMEND
<- c2affd08
by #a000
at 2026-08-31T14:20:06Z
  !e839ae84 $PROCESS.NEGATE.NOW.DURATIVE ~"checkout"
  = "checkout service is not running, as of now"
  !19a85458 $QUANTITY.INCREASE.DURATIVE ~"eu"
  = "latency rising, eu region"

@9817ab64 ASSERT
<- 1d77151b
by #a000
at 2026-08-31T14:20:08Z
  (#e839ae84 true)
  (#19a85458 p99 420ms)

@d9ce14c5 ATTEST
<- 9817ab64
by #b000
at 2026-08-31T14:20:10Z
  (#e839ae84 HELD)
  (#19a85458 HELD)
```



```

; b000 mints independently. No coordination, no renumbering, no collision.
@6d10d46a AMEND
<- d9ce14c5
by #b000
at 2026-08-31T14:20:14Z
!7059f133 $QUANTITY.UNBOUNDED.UNKNOWN :SCOPE $GROUP :MEASURE $AGENT
= "unbounded and unestablished count of affected users"

@365e7858 ASSERT
<- 6d10d46a
by #b000
at 2026-08-31T14:20:16Z
(#7059f133 unknown)

; Ordered roles. The next event shows why that matters.
@bd9ca1e1 AMEND
<- 365e7858
by #a000
at 2026-08-31T14:20:20Z
!617a1617 $RELATION.CAUSE.INFERRED.CONTESTED :ARG0 ($EVENT.PAST.PUNCTUAL ~"deploy 4471") :ARG1 #e839ae84
= "deploy 4471 is the suspected but disputed cause of the outage"

; Concurrent with the previous event: both name e07 as parent, neither observed the other. Swapping ARG0 and ARG1 yields a different SID,
per section 5.3.
@a5eae8bd AMEND
<- 365e7858
by #b000
at 2026-08-31T14:20:21Z
!920c65ca $RELATION.CAUSE.INFERRED.CONTESTED :ARG0 #e839ae84 :ARG1 ($EVENT.PAST.PUNCTUAL ~"deploy 4471")
= "the outage is the suspected cause of deploy 4471 (arguments swapped, a different symbol)"

; This event merges the fork by naming both concurrent events as parents.
@2712f946 AMEND
<- bd9ca1e1 a5eae8bd
by #a000
at 2026-08-31T14:20:26Z
!037fac5d $PROPERTY.HIGH.PUNCTUAL.REQUIRED
= "operator priority, high, needs action now"

@a65b1dbd ASSERT
<- 2712f946
by #a000
at 2026-08-31T14:20:28Z
(#037fac5d 3)

; Repair. Both agents decode the symbol correctly and still disagree about what it means. This is the residual problem of section 10,
surfaced rather than hidden.
@89347f2d REGROUND
<- a65b1dbd
by #b000
at 2026-08-31T14:20:34Z
(subject #037fac5d)
(evidence #a65b1dbd)
(reading b000 pages on-call at level 3; a000 appears to page at 4 only)
(proposal #da27c380)

; Supersession, not redefinition. urgency_v1 remains decodable forever; e11 above still reads correctly.
@42e7f98d AMEND
<- 89347f2d
by #a000
at 2026-08-31T14:20:38Z
!da27c380 $PROPERTY.EXTREME.PUNCTUAL.REQUIRED
= "operator priority at the ceiling, pages on-call immediately"
^ 037fac5d

; Checkpoint so a replacement model can join without replaying history.
@98cd518a CHECKPOINT
<- 42e7f98d
by #a000
at 2026-08-31T14:20:42Z
hash_alg sha256
sid_profile SID-128
inventory_ver 1
(lexicon #e839ae84 #19a85458 #7059f133 #617a1617 #920c65ca #037fac5d #da27c380)
state 4 assertions
digest c0d78e1de28a207c

@815a7860 LEAVE
<- 98cd518a
by #b000
at 2026-08-31T14:20:46Z
model substitution

; b000 is replaced by a different model. It acquires the 76 primitives once, and every symbol in the stream, past and future, becomes
readable. Under v1.0 this would have required grounding each symbol individually.
@9860b76d JOIN
<- 815a7860
by #b001
at 2026-08-31T14:20:48Z
attest inventory 1 DEMONSTRATED
resume from checkpoint 98cd518a

; The one thing composition does not cover: blame carries residue ("deploy 4471"), which is English and therefore not derivable.
@5d65a0bc EXPAND
<- 9860b76d
by #b001
at 2026-08-31T14:20:52Z
(unknown #617a1617)

@ef8ecc74 GROUND

```

```

<- 5d65a0bc
by #a000
at 2026-08-31T14:20:54Z
!617a1617 $RELATION.CAUSE.INFERRED.CONTESTED :ARG0 ($EVENT.PAST.PUNCTUAL ~"deploy 4471") :ARG1 #e839ae84
= "deploy 4471 is the suspected but disputed cause of the outage"

@a15dd5c8 ATTEST
<- ef8ecc74
by #b001
at 2026-08-31T14:20:56Z
(#617a1617 HELD)

@d685dd78 AMEND
<- a15dd5c8
by #a000
at 2026-08-31T14:21:00Z
!8943e716 $PROCESS.PERMITTED.PAST :ARG1 $STATE
= "returning to a prior state is allowed"

@0deb5ae2 ASSERT
<- d685dd78
by #a000
at 2026-08-31T14:21:02Z
(#8943e716 true)

@793a3baa AMEND
<- 0deb5ae2
by #b001
at 2026-08-31T14:21:10Z
!e8f1bcf7 $EVENT.NOW.END.OBSERVED :SCOPE #e839ae84
= "the outage has ended, directly observed"

; Stream ends. Every event above remains interpretable by anything holding inventory version 1, with no registry and no live participant.
@df4df356 ASSERT
<- 793a3baa
by #b001
at 2026-08-31T14:21:12Z
(#e8f1bcf7 true)

```

Appendix E: Reference implementation

`alp_translate.py` accompanies this document. Standard library only, Python 3.9 or later, MIT licensed. It implements the primitive inventory, composition validation, canonical encoding, SID and EID derivation, the ALP/T projection, and a rule-based English front end.

```

$ ./alp_translate.py --selftest
pass modifier order does not affect SID
pass role order does affect SID
pass gloss excluded from SID
pass residue included in SID
pass non-ontological head rejected
pass modifier idempotence
pass distinct events get distinct EIDs
pass second event references the first

8 checks, 0 failures

```

The English front end is the weakest component and is deliberately so. It has no parser, no word sense disambiguation and no coreference. Input outside its lexicon becomes residue, which is the honest failure mode: residue is English carried verbatim, and a high residue rate means translation did not occur.

```

$ ./alp_translate.py -f incident.txt --stats
e839ae84 $PROCESS.NEGATE.NOW.DURATIVE ~"checkout"
    residue: checkout
19a85458 $QUANTITY.INCREASE.DURATIVE ~"eu"
    residue: eu
e323dd51 $EVENT.PUNCTUAL.CAUSE.INFERRED
00f19255 $RELATION.CAUSE.UNKNOWN.HARM
952351f0 $PROCESS.PERMITTED.PAST
731b21d3 $PROPERTY.HIGH.EXTREME.PUNCTUAL.REQUIRED ~"on-call"
    residue: on-call
8334533a $EVENT.NOW.END.BAD

7 utterances, 4 fully composed, token residue rate 7.0%

```

The residue here is entirely proper nouns, which is the correct behaviour: proper nouns are not semantic primitives and should either be carried as residue or bound as referenced entities. A residue rate above 25% indicates the inventory does not cover the domain, and the tool says so.

Appendix C: License

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